

01-June'24

MASS SPECTROMETRY...

• Definition:

Mass spectrometry is technique for the analysis of substance in which a molecule is subjected to bombardment to cause ionization to give rise to ions in gas phase.

• Principle:

Separation of ionized gas phase atoms, molecules and fragments by the difference in their mass to charge ratio using mass spectrometer.

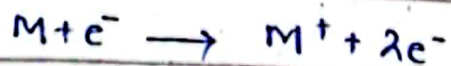
$$m/z$$

m = mass expressed in unified atomic mass unit

z = No. of charges on ions.

Formation of Ions:

Bombardment of sample with electrons.



Molecular ion on fragmentation is converted into ion of lower m/z .

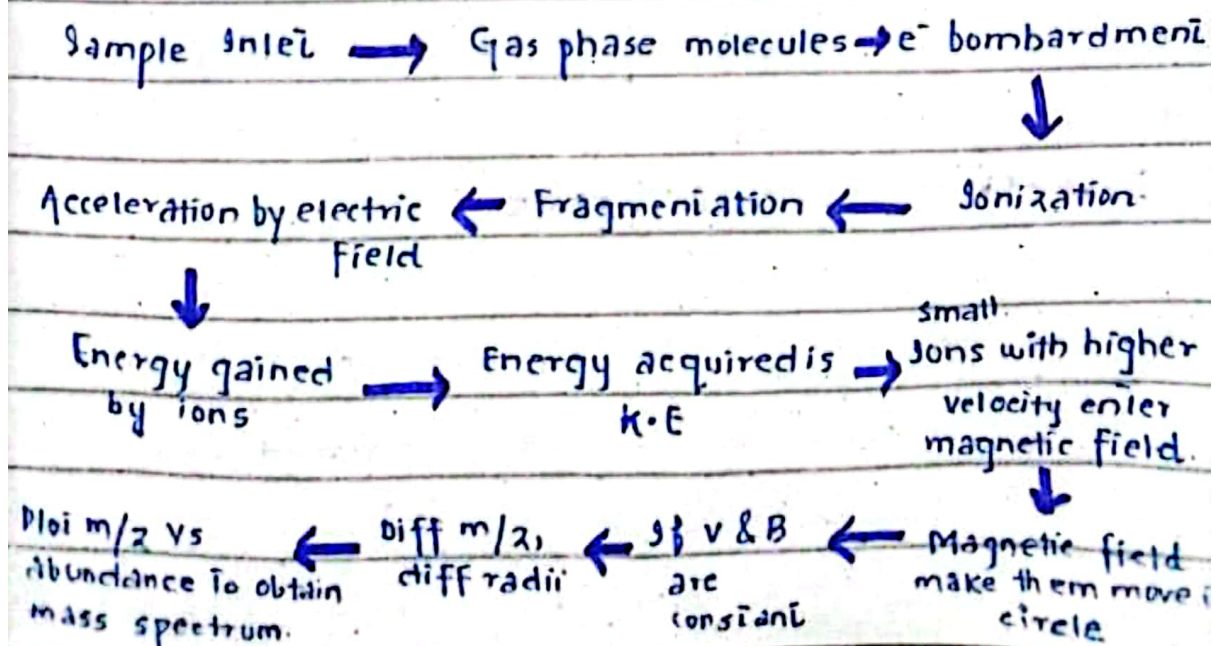
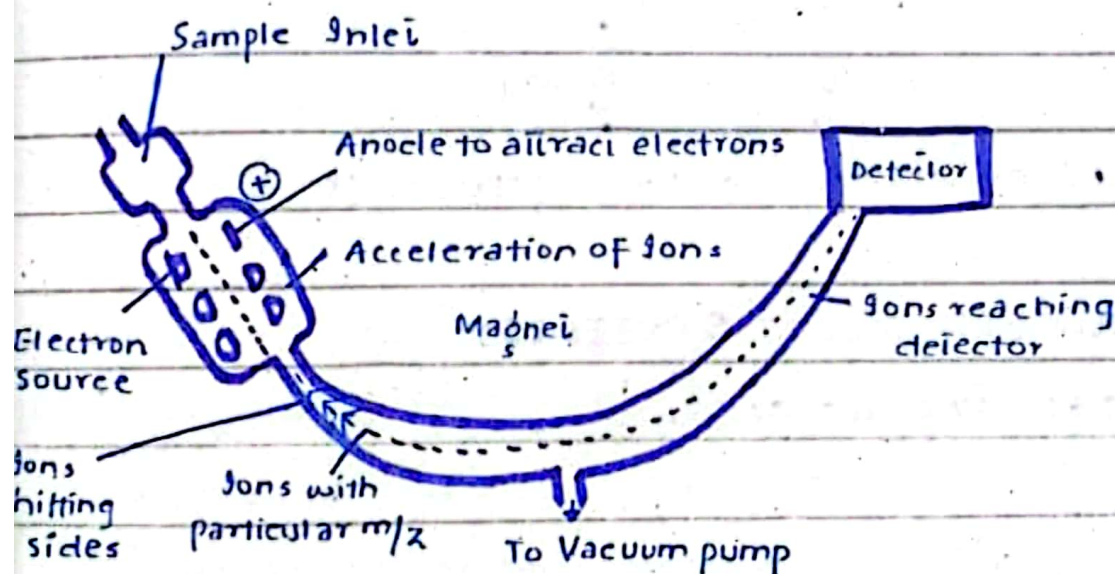
• Explanation:

Technique for creating gas phase ions from molecules, separating them according to

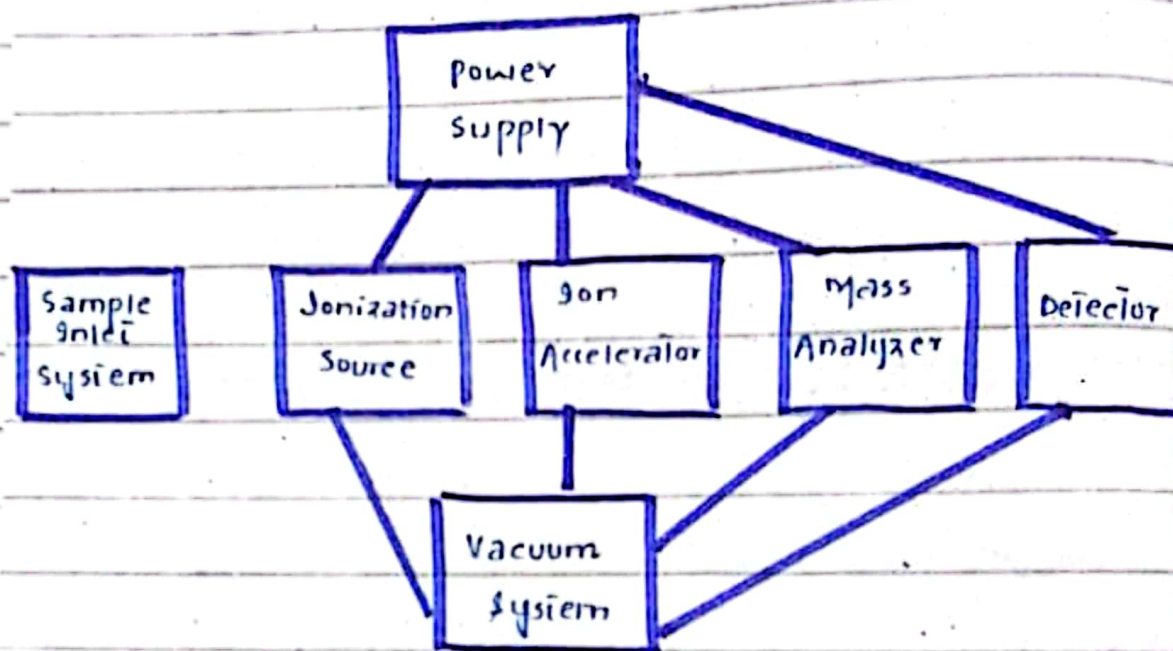
mass to charge value & measuring relative abundance.
Used to determine:

- Mass of molecules & ions.
- Molecular structure of organic & inorganic compounds.
- Identification and quantification of organic molecules and biomolecules.

MAGNETIC SECTOR MASS SPECTROMETER.



INSTRUMENTATION:



a Block diagram of mass spectrometer.

SAMPLE INLET SYSTEMS:

1) Gas Expansion:

Useful for gases and liquids with high vapour pressure. Sample is placed in ^{*}(glass capillary and inserted into ionization source) heated vessel and allowed to expand then leaked into ionization source.

2) Direct Insertion Probe:

For liquids with high BP and solids with high VP. Sample is inserted into glass capillary then inserted into

ionization source. Spectrometer contamination chances are high.

3) Direct Exposure Probe:

Sample is dissolved in solvent and a drop of solution is placed on probe, solvent evaporated, thin layer of sample is left on probe which is then inserted into ionization source. Spectrometer contamination chances are low.

≡ IONIZATION SOURCES:

1) Electron Ionization:

Electrons emitted from metal filament are accelerated by potential of 50-100 V towards anode, hard ionization source.

2) Chemical Ionization:

Large excess of reagent gas such as methane, isobutane, soft ionization source.

≡ MASS ANALYZER:

It is used to differentiate ions according to their mass to charge ratio. It may be single focusing or double focusing.

Magnetic & Electric Sector Instrument:

Magnetic sector instrument is used to separate ions according to their mass. Electric sector

Instrument is used to separate ions according to their m/e . If it combines both sectors it is called double focusing mass spectrometer.

DETECTORS:

Electron Multiplier (EM):

It is very sensitive

and give fast response, based on dynodes.

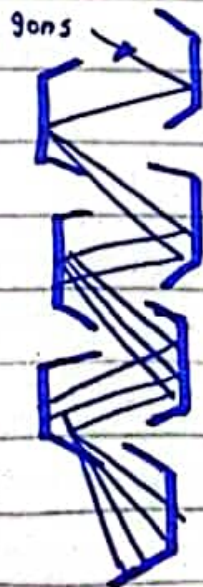
1) Discrete Dynode EM:

Use series of 12-24

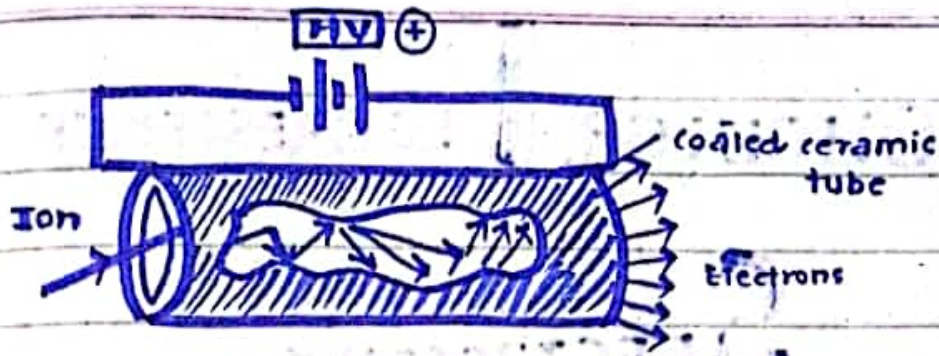
dynodes, each biased more $+ve$ than preceding dynode. Collision releases electrons from dynode surface which arrive at collector (amplified current).

2) Continuous Dynode EM:

Uses continuous glass tube coated inside with a conductive surface.

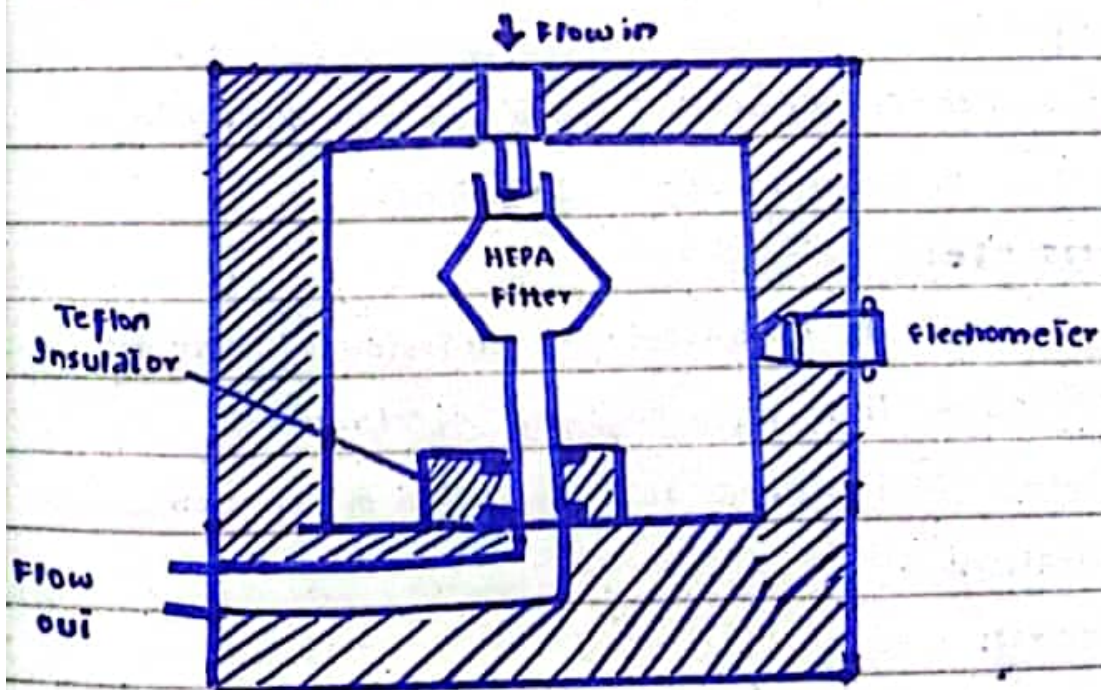


A Schematic diag of discrete dynode EM



Faraday Cup:

It consists of metal or C cup used to store charge, resulting in measured & amplified. There is decreased loss of electrons due to cup shape. Long response time limits its utility.



• schematic diag of Faraday Cup